

Udit Desai

PRINCIPAL SOFTWARE ENGINEER

Sydney, NSW, Australia

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"Walk the talk. Turn over every stone. Keep it human."

Summary

Principal Software Engineer with 10+ years building publishing platforms at scale. Building the tools that empower newsrooms behind [Sydney Morning Herald](#), [Australian Financial Review](#) & [The Age](#) to deliver quality journalism efficiently.

Currently, shaping multi-team technical direction for AI agent engineering at Nine — building a custom GitHub Copilot plugin that encodes Nine's own engineering standards, helping teams ship from idea to production without cutting corners.

Architecting Nine's Publishing Platform — connecting a React block editor to backend APIs (Go/GraphQL/Kafka) via a custom 2-way transform layer. Replaced a legacy PHP monolith to unlock cross-team contribution and reduce dependency on a single backend language.

Earned the Tech CTO Award ("Turn Over Every Stone") for persistence in solving hard problems. I also mentor engineers, help them grow in their careers, and build consensus across engineering and executive stakeholders.

Skills

Core Expertise

- CMS & Publishing platforms
- Editorial Systems
- Block Editors
- Microservices Architecture
- API Design
- Testing Strategies
- AI Agent Engineering: GitHub Copilot, Claude Code, OpenCode. Skills & Hooks

Technical Stack

Languages

TypeScript, JavaScript, PHP, Go, ShellScript, Ruby

Frontend

React, Gutenberg, NextJS, Redux, Apollo GraphQL, BackboneJS, MeteorJS

Backend

Node.js, Express, WordPress, GraphQL Server, gRPC, Kafka

Database

PostgreSQL, MySQL, MongoDB

Cloud & DevOps

AWS, Docker, Version Control (Git)

Testing

Jest, React Testing Library, Playwright, Cypress, PHPUnit

Work Experience

Nine Publishing

PRINCIPAL SOFTWARE ENGINEER

[Sydney, Australia](#)

Jan 2025 – Present

- Advising senior leadership on architectural decisions and shaping technical directions on multi-team initiatives.
- Developing AI agent engineering practices and tooling across the organization.
 - Building custom GitHub Copilot plugin with skills, knowledge base, and references encoding Nine's coding standards and architecture practices.
 - Helping with the adoption and rollout across teams. Teams use it to ship features from ideation to prototype to production faster.
- Mentoring engineers across teams through code reviews and knowledge sharing. I become the multiplier of engineering skills and capabilities across teams.

Nine Publishing

SENIOR SOFTWARE ENGINEER

[Sydney, Australia](#)

Feb 2020 – Jan 2025

- Architected a lean CMS layer connecting Gutenberg to internal APIs (Go/GraphQL/Kafka), replacing a legacy WordPress monolith.
- Operated as acting Engineering Manager, Delivery Manager, Product Manager during various manager transitions.

Nine Publishing

SOFTWARE ENGINEER

[Sydney, Australia](#)

Apr 2018 – Feb 2020

- Led development of Ink Authoring — CMS module of Nine's Publishing platform.
- Drove adoption of React and TypeScript across the team.

Santa Clara University

RESEARCH ASSISTANT

[Santa Clara, USA](#)

Jan 2016 – Jan 2017

- Developed HOME application v1.0 for Monterey City council.
- Integrated HOME app with HMIS hadoop cluster by implementing OAuth 2.0 strategy.
- Implementing VI-SPDAT scoring algorithm to verify 10,000+ homeless people for housing
- Verified 10,000+ homeless people for housing placement via HMIS Hadoop cluster (OAuth 2.0).
- Reduced testing time 70% via TravisCI/Mocha automation.

- Increased revenues for clients with 20% by developing 10 websites from scratch & developing WordPress themes & plugins.
- Increased rtCamp's CRM performance with 25% by leading the development team for rtBiz - CRM tool.

Technical Leadership

Nine Publishing

Sydney, Australia

NOVEL CMS ARCHITECTURE — GUTENBERG + MICROSERVICES BRIDGE

2024 – Present

- Problem: A WordPress monolith blocked cross-team contribution. Every code change required PHP knowledge, and newsroom editors needed better tools. But we couldn't move fast enough on legacy codebase.
- Solution: Re-build of the CMS layer to connect a React block editor to our internal APIs (Go/GraphQL/Kafka) — bypassing WordPress entirely for content storage.
 - 2-way transform layer between Gutenberg blocks and internal “Ink Blocks.”
 - ★ Write path: Gutenberg block array → Ink Blocks → GraphQL → gRPC → Kafka.
 - ★ Read path: reverse. Required reverse-engineering Gutenberg's internal data flow.
 - Reused select @wordpress/ packages (@wordpress/blocks, @wordpress/data, @wordpress/rich-text) with custom equivalents for the rest.
 - gRPC with proto contracts for compile-time schema enforcement. Kafka decouples authoring from downstream publication.
 - Built PoC first, then showed it to leadership. Near production launch.
- Result: Increased development velocity. Other engineering teams now contribute using React/TypeScript. Reduced reliance on PHP.

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Sydney, Australia

ENGINEERING EXCELLENCE — AI, TOOLING & CAPABILITY BUILDING

2023 – Present

- Built AI agents for Git commit message generation and unit test writing
 - Picked these two because they're high-impact and low-risk for AI augmentation.
- Introduced Orbstack (Docker Desktop replacement), RenovateBot for dependency updates, and post-deploy Playwright smoke tests across the team.
- Coaching & Mentoring: Engineers on the team grew to handle complex frontend work independently.
 - Built a culture of knowledge sharing and continuous learning.
 - Leadership style: open-ended questions over directives; explain the “why”; show vulnerability to lower authority barriers; rubber-ducking with screen share.
- Ran Engineers' Round Table forum and Deep Dive technical sessions. Established patterns and conventions that multiplied team velocity.

Nine Publishing

Sydney, Australia

BRIGHTCOVE DECOMMISSION – PROBLEM → ADVOCACY → PoC → INFLUENCE → EXECUTION → QUALITY → LAUNCH

2026

- Problem: The legacy Brightcove plugin kept causing production incidents — external vendor API failures were taking down the app.
- Solution: Built a replacement using React components already in the new architecture. Pitched the PoC to leadership.
- Execution: Offered to lead the rollout myself since the team was busy with other priorities.
- Quality: Conducted bug bash, wrote test cases document to help the team, fixed critical bugs before launch.
- Result: Zero production incidents from this source since launch.

Notable Projects

Nine Publishing Hackathon

Sydney, Australia

AI GHOST TEXT

2025

- Brought AI autocomplete into Gutenberg via Alt+Tab shortcut. Journalists never leave the editor to use external AI tools.
- Presented to senior leadership; production committed for 2026.

Nine Publishing Hackathon

Sydney, Australia

INK AI SEO HEADLINES

2023

- Built generative AI tool to suggest better SEO headlines. Goal: 50% decrease in low-engagement stories. Now used daily by newsrooms.
- Won **Outstanding Team Work** award for excellent collaboration between product, engineering, design, and editorial teams.

Nine Publishing

Sydney, Australia

EDITORIAL NOTES

2023 – 2024

- Owned full technical arc: requirements, architecture, frontend spike, GoLang backend, GraphQL API, integration testing.
- Coached team members to take charge of frontend work. Supported sister team for print workflow implementation.

Awards & Recognition

COMPANY AWARDS

Aug 2023	Tech CTO Award — “Turn Over Every Stone” , for persistence in solving hard problems, Nine Publishing	<i>Sydney, Australia</i>
Nov 2023	INMA Award — Ink Authoring , International News Media Association	
Jul 2023	Ink AI SEO Headlines —Outstanding Team Work , Nine Hackathon	<i>Sydney, Australia</i>

HACKATHON AWARDS

Apr 2017	Best use of AWS , BroncoHack, Github	<i>Santa Clara, USA</i>
Jan 2017	Best use of Microsoft Azure , HackUCSC, Github	<i>UC Santa Cruz, USA</i>
Nov 2016	Top 6 Finalists , HackRice, Github	<i>Rice University, USA</i>
Sep 2016	Best WordPress Data Hack , TechCrunch Disrupt , Github	<i>San Francisco, USA</i>
Jun 2016	Best User Voice Experience , HackDavis, Github	<i>UC Davis, USA</i>

Other Work

OPEN SOURCE

WordPress / Gutenberg

WORDPRESS CONTRIBUTOR	<i>2014 – 2021</i>
- Contributions in Gutenberg - Contributions in WordPress Core - Other WordPress Plugins	

FREELANCE WORK

10up	<i>Remote / EMEA (UK)</i>
WEB ENGINEER	<i>Jun 2022 – Sep 2023</i>
Moxie Media Group	<i>Remote / USA</i>
LEAD ENGINEER	<i>Dec 2014 – Aug 2015</i>
XWP.co Pty. Ltd.	<i>Remote / Australia</i>
WORDPRESS ENGINEER	<i>Feb 2014 – Aug 2015</i>

Education

Santa Clara University	<i>Santa Clara, CA</i>
M.S. COMPUTER SCIENCE	<i>June 2017</i>
DA-IICT	<i>Gandhinagar, India</i>
M.S. INFORMATION TECHNOLOGY	<i>May 2013</i>
University of Pune	<i>Pune, India</i>
B.S. COMPUTER SCIENCE	<i>June 2011</i>